Lesson Planning

Complete Lesson Plan Book

Class: FSc Pre-Engineering · Class 11

Subject: Mathematics

Medium: English

Academic Year 2026-27

This book is prepared as a teaching guide for teachers.

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Unit 1: Quadratic Equations: Solving by Factorisation

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Unit 1: Quadratic Equations: Solving by Factorisation

Learning Objectives

• Students will factor quadratic expressions by splitting the middle term.

• Students will solve quadratic equations using the zero-product rule.

• Students will translate a simple word problem into a quadratic equation.

Key Concepts

• A quadratic equation has the form ax squared plus bx plus c equals 0, with a not equal to zero.

• Factorisation rewrites the quadratic as a product of two linear factors.

• The zero-product rule says that if a product is zero, at least one factor must be zero.

• Splitting the middle term helps factor quadratics where the leading coefficient is 1.

Teaching Methodology (Step by Step)

1. Recall linear equations and ask what changes when x is squared.
2. Factor x squared plus 5x plus 6 together, finding two numbers with product 6 and sum 5.
3. Introduce the zero-product rule: if (x minus 2)(x minus 3) equals 0, then x equals 2 or x equals 3.
4. Work a second example where the leading coefficient is not 1.
5. Students solve three equations alone, then check answers by substitution.
6. Close with the rectangle word problem solved step by step on the board.

Classroom Activities

• Factor hunt: pairs race to factor three quadratics and check each other by expanding.

• Root check: students substitute each root back into the original equation to prove it works.

• Real rectangle: a garden problem where length and width come from a quadratic equation.

Assessment Questions

1. Solve x squared minus 5x plus 6 equals 0 by factorisation.
2. Why must one factor equal zero for the zero-product rule to work?
3. A rectangle has area 24 square units and length 2 units more than its width. Set up the quadratic equation.

Homework

• Solve five quadratic equations by factorisation, showing the middle-term split for each.

• Write one word problem of your own that leads to a quadratic equation and solve it.

Time Allocation

40 minutes